

Task 1 – Packages: Organizing Java Code

Description:

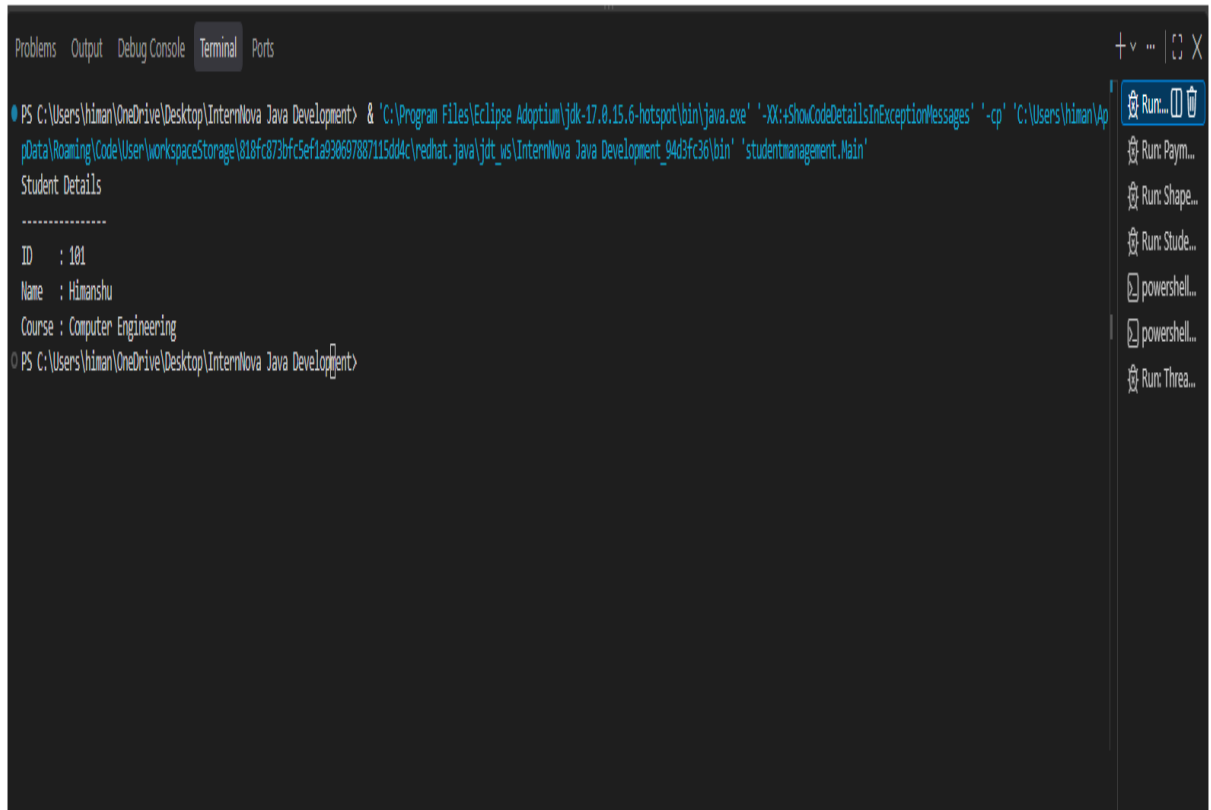
A package in Java is used to organize related classes and avoid naming conflicts. In this task, a custom package named studentmanagement is created.

The Student class is placed inside this package and contains student ID, name, and course. The Student class is then imported into another Java class using the import keyword. An object of the Student class is created to display student information.

Concepts Used:

- Package
- package keyword
- import keyword
- Class and Object
- Data members
- Constructor

TASK 1 OUTPUT:



```
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development> & 'C:\Program Files\Eclipse Adoptium\jdk-17.0.15.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\himan\AppData\Roaming\Code\User\workspaceStorage\818fc873bfc5ef1a930697887115d04c\redhat_java\jdk_ws\InternNova Java Development_9403fc36\bin' 'studentmanagement.Main'
```

Student Details

ID : 101

Name : Himanshu

Course : Computer Engineering

```
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development>
```

Task 2 – Interface: Payment System

Description:

An interface in Java defines a common set of methods that implementing classes must provide. In this task, a Payment interface is created with `pay()` and `showPaymentDetails()` methods. Two classes, `UPIPayment` and `CardPayment`, implement the interface. Both classes provide their own implementation of the payment methods. Objects of both classes are created to demonstrate how the same interface can be used for different types of payments.

Concepts Used:

- Interface
- implements keyword
- Method implementation
- Polymorphism
- Class and Object

TASK2 OUTPUT :

```
Suggestion [3,General]: The command ShapeDemo.java was not found, but does exist in the current location. Windows PowerShell does not load commands from the current location by default command, instead type: ".\ShapeDemo.java". See "get-help about_Command_Precedence" for more details.
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development\Abstract Class> javac ShapeDemo.java
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development\Abstract Class> java ShapeDemo
Calculating area...
Circle Area = 78.5

Calculating area...
Rectangle Area = 50
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development\Abstract Class> |
```

Task 3 – Abstract Classes: Shape Calculator

Description:

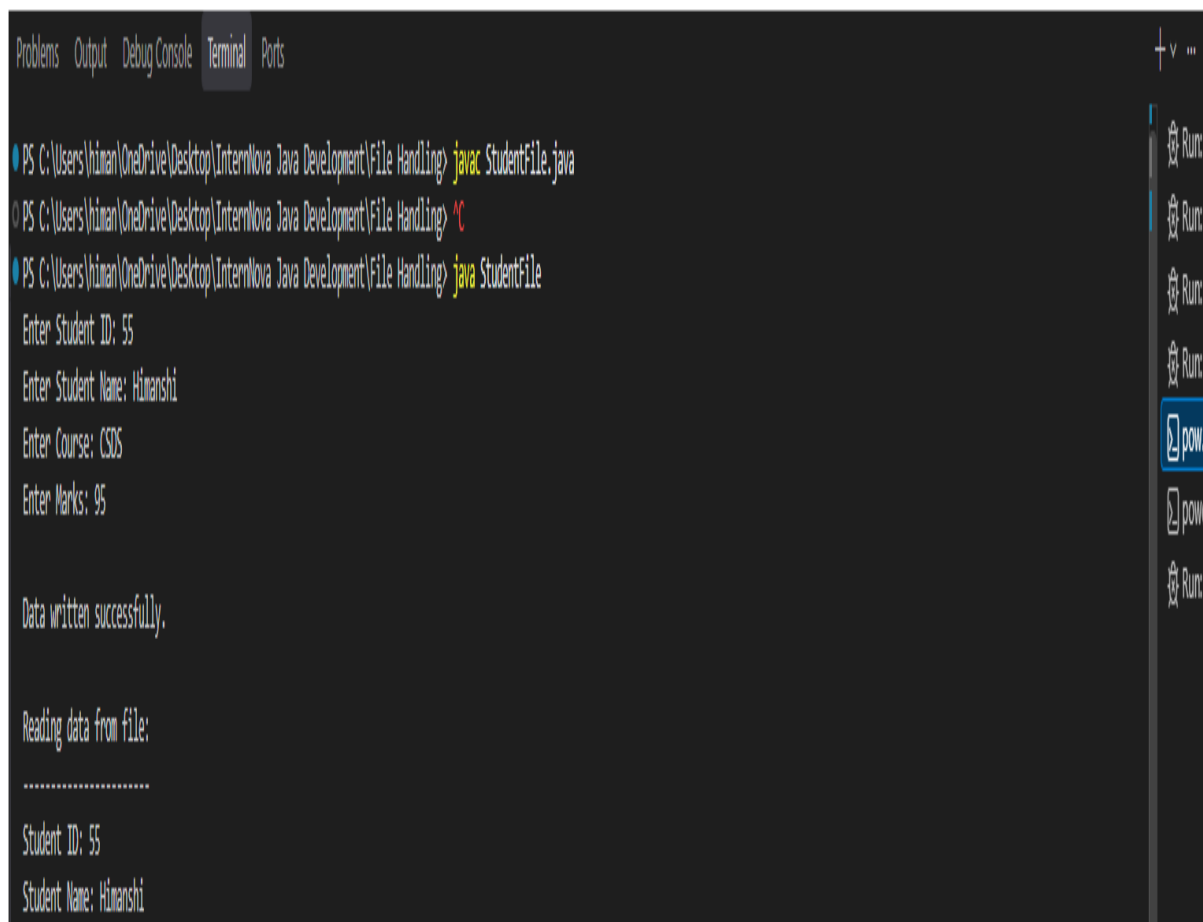
An abstract class is a class that can contain both abstract and normal methods. In this task, an abstract class named Shape is created. It contains an abstract method calculateArea() and a normal method displayMessage(). Two child classes, Circle and Rectangle, extend the Shape class and provide their own implementation of the calculateArea() method. Objects of both classes are created to calculate and display their respective areas.

Concepts Used:

- Abstract class
- abstract keyword

- Abstract method
- Normal method
- extends keyword
- Inheritance
- Method overriding

TASK 3 OUTPUT:



The screenshot shows an IDE terminal window with the following content:

```
Problems Output Debug Console Terminal Ports
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development\File Handling> javac StudentFile.java
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development\File Handling> ^C
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development\File Handling> java StudentFile
Enter Student ID: 55
Enter Student Name: Himanshi
Enter Course: CSOS
Enter Marks: 95

Data written successfully.

Reading data from file:
-----
Student ID: 55
Student Name: Himanshi
```

The terminal output demonstrates the execution of a Java program that reads student data from a file. The user enters the Student ID (55), Student Name (Himanshi), Course (CSOS), and Marks (95). The program then writes this data to a file and reads it back, displaying the stored information.

Task 4 – File Handling: Student Records

Description:

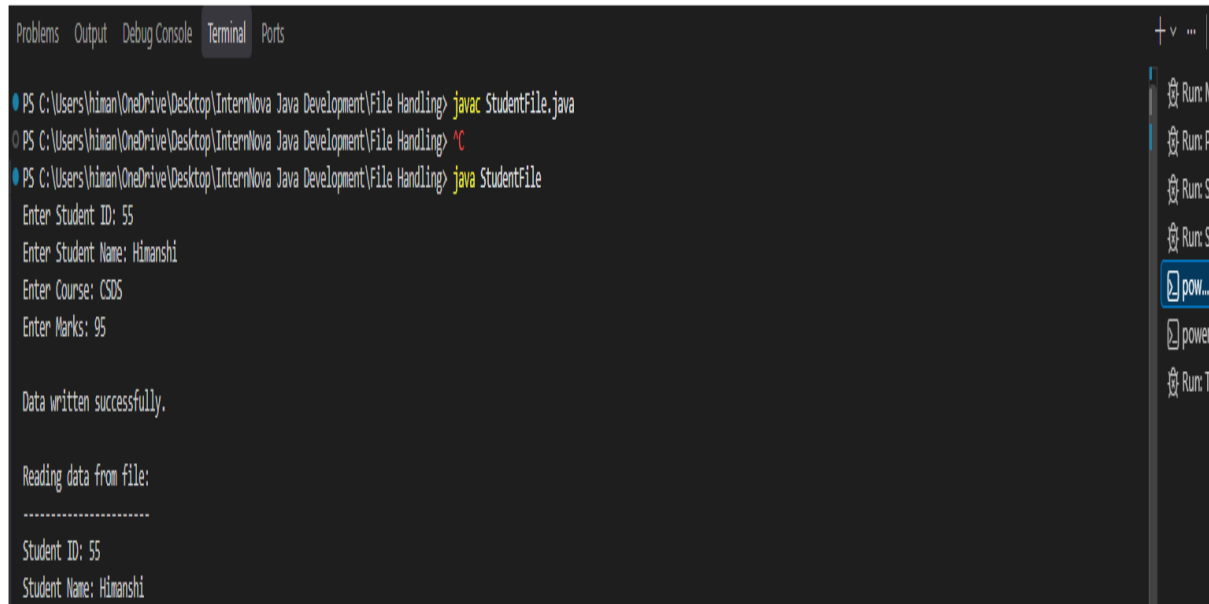
File handling in Java is used to store and retrieve data from files. In this task, a text file is created to store student information such as student ID, name, course, and marks. `FileWriter` is used to write the information into the file, while `FileReader` and `BufferedReader` are used to read the stored

information. Exception handling is also used to handle errors that may occur during file operations.

Concepts Used:

- File creation
- File writing
- File reading
- FileWriter
- FileReader
- BufferedReader
- Exception handling
- try-catch

TASK 4 OUTPUT :



The screenshot shows an IDE terminal window with the following content:

```
Problems Output Debug Console Terminal Ports
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development\File Handling> javac StudentFile.java
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development\File Handling> ^C
PS C:\Users\himan\OneDrive\Desktop\InternNova Java Development\File Handling> java StudentFile
Enter Student ID: 55
Enter Student Name: Himanshi
Enter Course: CSOS
Enter Marks: 95

Data written successfully.

Reading data from file:
-----
Student ID: 55
Student Name: Himanshi
```

The terminal output demonstrates the execution of a Java program that reads and writes student data to a file. The user enters the student ID (55), name (Himanshi), course (CSOS), and marks (95). The program confirms that the data was written successfully and then reads it back from the file, displaying the same information.

Task 5 – Multithreading Basics: Multiple Tasks

Description:

Multithreading allows multiple tasks to execute independently within a Java program. In this task, two threads are created using the Thread class. The first thread prints numbers from 1 to 10, while the second thread prints the message “**Learning Java Multithreading**” multiple times. Both threads are started using the start() method. The output demonstrates that both tasks can execute concurrently.

Concepts Used:

- Multithreading
- Thread class
- run() method
- start() method
- Thread.sleep()
- Concurrent execution

TASK 5 OUTPUT :

Learning Java Multithreading

Number: 2

Learning Java Multithreading

Number: 3

Learning Java Multithreading

Number: 4

Learning Java Multithreading

Number: 5

Learning Java Multithreading

Number: 6

Learning Java Multithreading

Number: 7

Learning Java Multithreading

Number: 8

Learning Java Multithreading

Number: 9

Learning Java Multithreading

Number: 10

PS C:\Users\human\OneDrive\Desktop\InternNova Java Development>

Run: Me

Run: Pa

Run: Sh

Run: Sh

powers

powers

Run...